Topic Breadth: Six Analytics Areas in Real Business Decisions

Each example states the analytical output and the decision it can support. Selected project topic: supervised machine learning / regression.

1. Supervised Machine Learning

A taxi fleet can learn from historical trips for which the actual trip duration is already known. Inputs available at pickup, including pickup time and trip endpoints when the destination is known, can be used to predict a continuous trip-duration target. Kaggle's NYC Taxi Trip Duration competition is a real supervised regression setting. [1]

> **Decision supported:** use predicted duration to plan dispatch, driver availability, and passenger ETA communication.

2. Unsupervised Machine Learning

A company can cluster customers with similar demographic or behavioral patterns without predefined labels. Hicham and Karim show customer segmentation as a decision-support and marketing application of clustering methods. [2]

> **Decision supported:** decide which customer groups should receive different promotions, messages, or product strategies.

3. Dimensionality Reduction

Telecom firms collect high-dimensional customer activity data. PCA can compress many correlated features into a smaller representation before clustering, while retaining important structure in the data. A telecom segmentation study directly evaluates PCA for this purpose. [3]

> **Decision supported:** simplify customer behavior data so analysts can build clearer segments and prioritize targeting or retention actions.

4. Natural Language Processing (NLP)

A bank can use NLP to classify incoming customer-support tickets. Bonechi et al. use 4,243 real chat-based requests from MPS Bank and compare text-classification methods for ticket categorization. [4]

> **Decision supported:** route requests to the correct support queue and allocate service resources more efficiently.

5. Semantic / Sentiment Analysis

An e-commerce retailer selling smartphones can use aspect-based sentiment analysis to separate opinions about specific features such as battery, camera, design, or service instead of relying on one overall rating. Research on annotated mobile-phone reviews and restaurant reviews shows how aspect sentiment reveals customer preferences and performance signals. [5][6]

> **Decision supported:** prioritize which product features or service problems should be improved first.

6. Social Media Analytics

A consumer brand launching a new product can analyze Facebook, Instagram, and other social-media conversations to identify preferences, complaints, and market signals. Peer-reviewed research connects these signals with customer understanding, product development, marketing investment, and broader business decisions. [7][8]

> **Decision supported:** decide which campaign messages, customer-service issues, or product changes deserve priority.

ONE-PAGE INFOGRAPHIC

Linear Regression: From Least Squares to Trip-Time Decisions

Why a transparent supervised regression model is a suitable first model for NYC taxi trip duration.

Origins and development

1805

Legendre
Published a least-squares formulation based on minimizing squared residuals. [9]

1809

Gauss
Presented a probabilistic formulation of least squares; later developed another least-mean-square interpretation. [9]

1886

Galton
Advanced the statistical idea of "regression" through work on hereditary stature and regression toward the mean. [10]

Today

Supervised ML
Ordinary least squares remains a standard regression method for continuous targets and minimizes residual sum of squares. [11]

Practical use

- **Transportation:** predict taxi ride duration from trip attributes. [1]
- **Finance:** regress future bond returns on forward rates. [12]
- **Forecasting:** linear predictive regressions are also used to study stock-return predictability. [13]

Value and impact

- Produces a numeric prediction that is easy to compare with a baseline.
- Coefficients make the model relatively transparent and interpretable.
- Useful as a fast benchmark before more complex models.
- **Caution:** statistical predictability does not automatically imply economic value or guaranteed gains. [13]

Connection to this project

- **Target:** trip duration in seconds.
- **Features tested:** Haversine distance and pickup hour.
- **Boundary:** `dropoff_datetime` excluded to avoid leakage.
- **Assumption:** destination coordinates are known at prediction time.
- **Metric:** RMSLE; lower is better. [1]

Information available at or before pickup > Feature engineering **distance_km + pickup_hour** > Linear regression on log1p(trip duration) > Trip-time estimate for planning and ETA support

Initial validation RMSLE (lower is better)



Continuous outcome

Trip duration is numeric, so regression directly matches the prediction target.

Transparent first benchmark

OLS gives an interpretable reference point before testing more complex models. [11]

Evidence over claims

Model value is judged on held-out RMSLE, not assumed savings or guaranteed business gains.

PROJECT PROPOSAL

New York City Taxi Trip Duration

Supervised regression proposal | Repository: github.com/fhu0914/MSBA265-NYC-Taxi-Final-Project | branch: main

Stakeholder problem

Trip-time uncertainty can affect taxi fleet planning, driver utilization, and passenger expectations. The hypothetical stakeholder is an NYC taxi fleet operations or urban mobility planning team that needs a transparent estimate at pickup time.

Focused question

Using only information available at or before pickup, how accurately can taxi trip duration be predicted?

The project treats this as a supervised regression problem with a continuous target.

Intended use

Support trip-time estimation and operational planning at pickup. The model is not intended as an official navigation system or for safety-critical decisions.

Dataset / source

Kaggle's NYC Taxi Trip Duration competition uses sampled and cleaned 2016 NYC Yellow Cab trip records originally published by the NYC Taxi and Limousine Commission. The training set contains 1,458,644 trips. [1]

Proposed method

Audit the data, engineer Haversine trip distance and pickup-time features, model *log1p(trip_duration)*, use an 80/20 train-validation split with random_state=42, compare a training-median baseline with ordinary least squares linear regression, and expand features only after a leakage check.

Success measure

Root Mean Squared Logarithmic Error (RMSLE), the official competition evaluation metric. A successful model should improve on the baseline while remaining interpretable and using only information realistically available at prediction time. [1]

Prediction-time boundary: `dropoff_datetime` is explicitly excluded because it is only known after the trip ends and would leak the target. The project assumes the passenger's destination coordinates are known at prediction time, which makes route-distance estimation permissible for this defined use case.

AI use disclosure: AI tools were used to assist with code generation, debugging, documentation, and explanations of technical concepts. The code and outputs were reviewed, tested, and revised by the student; AI was used as a support tool rather than a substitute for understanding the workflow.

Data Audit, Feasibility, and Early Evidence

Results reproduced from the project workflow included in this submission.

Training rows	Missing values	Median trip duration	Median Haversine distance
1,458,644	**0**	**662 s**	**2.094 km**
11 original columns	across the initial dataset audit	training-median baseline value	derived from pickup/dropoff coordinates

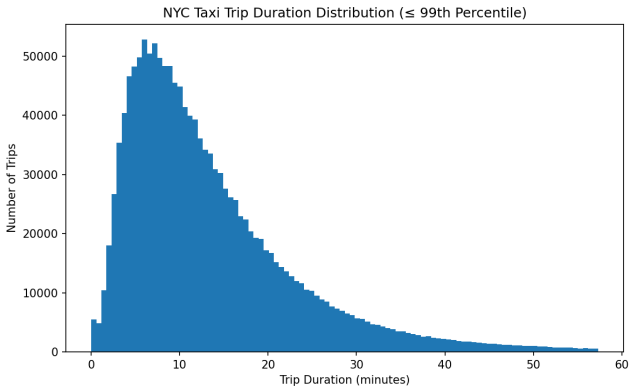


Figure 1. Raw trip-duration distribution displayed through the 99th percentile for readability. The strong right tail motivates a documented outlier audit and modeling `log1p(trip_duration)`; the display cutoff does not remove observations from the modeling data.

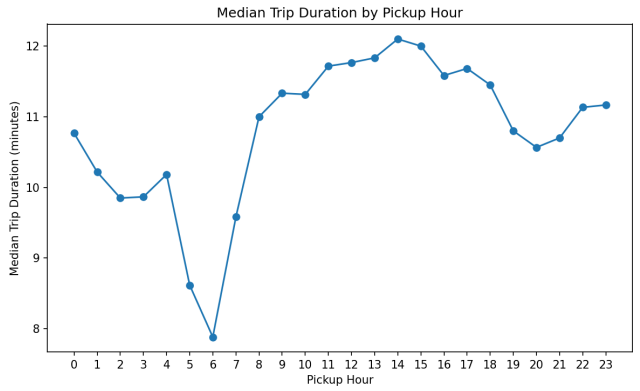


Figure 2. Median trip duration by pickup hour. Median duration ranges from about 7.9 minutes at 6:00 to about 12.1 minutes at 14:00. The nonlinear shape supports testing pickup time, while the small RMSLE improvement from 0.6388 to 0.6376 suggests that treating hour as one simple numeric feature captures only limited incremental signal.

Validation RMSLE

Lower values indicate better validation performance. [1]

Median baseline	**0.7971**
Distance only	**0.6388**
Distance + hour	**0.6376**

Interpretation: distance provides the large first improvement; pickup hour adds a smaller incremental gain.

What the evidence says

- **Target shape:** the visible right tail supports using a log-scale target while auditing, rather than automatically deleting, extreme observations.
- **Time pattern:** the hourly medians show a clear nonlinear daily pattern, with the shortest trips around 6:00 and the longest around 14:00. The small RMSLE gain from adding raw pickup hour indicates that the current linear encoding does not capture much of that pattern.
- **Extreme values:** maximum trip duration is 3,526,282 seconds and maximum derived distance is about 1,240.9 km; both require investigation before any filtering rule.
- **Next step:** add leakage-safe pickup-time features, audit extremes, and compare stronger models using the same held-out validation framework.

Brief interpretation: The initial models clearly beat the training-median baseline, so the project is feasible. The right-skewed target supports log-scale modeling, and the hourly pattern supports time-of-day as a candidate feature. However, the very small RMSLE improvement from adding raw pickup hour shows that visible EDA structure does not automatically translate into a large linear-model gain.

CREDIBLE CITATIONS

References

Peer-reviewed papers supplied for this assignment are combined with primary or official sources used for project data, method history, and model definition.

[1] Risdal, M. (2017). *New York City Taxi Trip Duration*. Kaggle. https://www.kaggle.com/competitions/nyc-taxi-trip-duration

[2] Hicham, N., & Karim, S. (2022). Analysis of unsupervised machine learning techniques for an efficient customer segmentation using clustering ensemble and spectral clustering. *International Journal of Advanced Computer Science and Applications, 13*(10). https://doi.org/10.14569/IJACSA.2022.0131016

[3] Alkhayrat, M., Aljnidi, M., & Aljoumaa, K. (2020). A comparative dimensionality reduction study in telecom customer segmentation using deep learning and PCA. *Journal of Big Data, 7*, 9. https://doi.org/10.1186/s40537-020-0286-0

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[10] Galton, F. (1886). Regression towards mediocrity in hereditary stature. *Journal of the Anthropological Institute, 15*, 246-263. https://doi.org/10.2307/2841583

[11] scikit-learn developers. (n.d.). *LinearRegression documentation*. https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

[12] Cochrane, J. H., & Piazzesi, M. (2005). Bond risk premia. *American Economic Review, 95*(1), 138-160. https://doi.org/10.1257/0002828053828581

[13] Giot, P., & Petitjean, M. (2011). On the statistical and economic performance of stock return predictive regression models: An international perspective. *Quantitative Finance, 11*(2), 175-193. https://doi.org/10.1080/14697680903468971

Source integrity note: Financial and impact statements are descriptive examples from the cited literature; this submission does not claim guaranteed savings, profits, or social impact from linear regression. DOI-based academic references were cross-checked against publisher or institutional records. Project-specific RMSLE values and data-audit statistics are outputs from the student's reproducible analysis rather than external claims.